# Functional Annotation Report: Non-Homologous End Joining Protein Ku (gene *ku*, PP_3255, UniProt Q88HU8) in *Pseudomonas putida* KT2440

## Summary


The gene **ku** (locus **PP_3255**; UniProt **Q88HU8**) of *Pseudomonas putida* KT2440 encodes the **Non-Homologous End Joining (NHEJ) protein Ku**, the DNA-double-strand-break (DSB) end-binding component of the bacterial NHEJ DNA-repair pathway. Ku is **not an enzyme**; it is a sequence-independent DNA-end-recognition and end-bridging protein. Its identity is unambiguous: the UniProt record assigns it to the prokaryotic Ku family (Pfam **PF02735** "Ku"; InterPro **IPR009187** "Prok_Ku"; **IPR006164** Ku70/Ku80 DNA-binding core; **IPR016194** SPOC-like C-terminal domain), and *P. putida* KT2440 is one of the minority of bacteria that encodes both Ku and its obligate catalytic partner **DNA ligase D (LigD)** — the two proteins that constitute the complete, self-sufficient bacterial NHEJ machine. This report therefore concerns the correct target: the bacterial (prokaryotic) Ku, which functions as a **homodimer**, in contrast to the eukaryotic Ku70/Ku80 heterodimer.

Mechanistically, Ku performs the first three of the five canonical NHEJ phases — **end sensing, end protection, and end tethering**. It recognizes and clamps blunt or near-blunt broken dsDNA ends without regard to sequence, shields those ends from nucleolytic (exonuclease) degradation, threads inward along the DNA duplex, and oligomerizes into higher-order arrays that bridge and synapse the two broken ends. Through a defined C-terminal subregion, Ku then recruits and stimulates LigD, handing off the held ends to the catalytic machine (LigD supplies polymerase, 3′-phosphoesterase, and ATP-dependent ligase activities) that processes and reseals the break. Ku is the **rate-limiting factor** of the pathway and the structural anchor on which the entire end-joining apparatus is built.

Biologically, Ku-dependent NHEJ provides *P. putida* with a **template-independent, RecA-independent, intrinsically mutagenic** route to repair chromosomal DSBs. This pathway is most important when a homologous sister template is unavailable — for example in stationary

phase or dormancy — and it operates in the cytoplasm/nucleoid directly at break sites on chromosomal DNA. In *P. putida* KT2440 specifically, the presence of Ku+LigD makes the strain a valuable genome-engineering chassis, because Ku-dependent NHEJ shapes the mutational outcomes of programmable DSBs introduced by CRISPR-Cas9 and I-SceI editing tools.

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## Key Findings

### Finding 1 — Ku (PP_3255, Q88HU8) is the DNA-end-binding component of prokaryotic NHEJ in *P. putida*, acting with LigD

The identity of the target is verified at the sequence and family level. UniProt **Q88HU8** corresponds to gene *ku* / locus **PP_3255** and is assigned to the prokaryotic Ku family through Pfam **PF02735** ("Ku"), InterPro **IPR009187** ("Prok_Ku"), and the Ku70/Ku80 DNA-binding core domain **IPR006164**. Crucially, *P. putida* KT2440 is documented to encode **both** Ku and LigD — the two-component minimal NHEJ system — a feature that distinguishes it from most bacteria. This pathway repairs Cas9-generated double-strand breaks in vivo in *P. putida* (PMID: 36475478): *"Unlike most bacteria, P. putida KT2440 encodes the Ku and LigD proteins involved in Non-Homologous End Joining (NHEJ)."*

A defining structural feature separating the bacterial protein from its eukaryotic namesake is quaternary organization: **bacterial Ku functions as a homodimer**, whereas the eukaryotic version is a Ku70/Ku80 heterodimer (PMID: 41118517): *"bacterial NHEJ operates with a simpler toolkit comprising a Ku homodimer and the multifunctional LigD."* This homodimeric architecture and the reduced "toolkit" define the mechanistic context in which the *P. putida* protein operates.

### Finding 2 — Ku binds DNA ends, bridges/synapses the two broken ends, threads inward, and recruits LigD via its C-terminus

Structural and biochemical studies of bacterial Ku homologs converge on a consistent mechanistic picture. Ku is a **ring/clamp-structured protein** that binds DNA ends and then recruits downstream factors that access the ends by **threading of Ku inward** along the DNA (PMID: 26961308): *"The ring structured eukaryotic Ku binds DNA ends and recruits other factors which can access DNA ends through the threading of Ku inward the DNA."* The same study dissects the C-terminus and assigns discrete mechanistic roles to its subregions: *"the minimal C-*

*terminus is required for the Ku-LigD interaction, whereas the basic extension controls the threading and DNA bridging abilities of Ku."* Thus the **minimal C-terminal region mediates LigD recruitment**, while the **basic C-terminal extension governs threading and DNA bridging**.

Two recent cryo-EM studies provide direct structural confirmation of the synapsis function. Cryo-EM of *Mycobacterium tuberculosis* Ku shows it forms an **extended proteo-filament** upon DNA binding, with the C-terminus regulating DNA loading and facilitating LigD recruitment (PMID: 41298423): *"the C-terminus of Ku regulates DNA binding and loading and facilitates subsequent recruitment of LigD."* Independently, a 2.74 Å cryo-EM structure of *Bacillus subtilis* Ku captured **two blunt DNA ends bridged by Ku alone**, supporting a model in which oligomeric arrays of Ku homodimers physically synapse the break (PMID: 41118517): *"oligomeric arrays of Ku homodimers bridge and stabilize two DNA ends, facilitating efficient DSB repair."* Together these findings establish that Ku itself synapses the two broken ends prior to ligation and controls the hand-off to the catalytic step.

## Finding 3 — Ku-dependent NHEJ repairs DSBs in *P. putida* and shapes error-prone (mutagenic) end-joining outcomes

In *P. putida* KT2440, genetic manipulation of the NHEJ machinery has direct, measurable consequences on DSB repair. Removing or overproducing Ku/LigD alters the spectrum of mutations generated upon repair of Cas9-mediated DSBs, and the pathway carries intrinsic mutagenic potential (PMID: 36475478): *"This pathway of repair of double-strand breaks (DSBs) in DNA has an intrinsic mutagenic potential."* This establishes a bona fide, non-templated DSB-repair activity operating in the living cell and demonstrates that Ku status is causally connected to repair outcomes.

The central importance of Ku is reinforced by cross-pathway genetics in the related pseudomonad *P. aeruginosa*, where deletion analysis found **RecA and Ku (but not LigD) conditionally essential**, revealing that *Pseudomonas* can perform even **LigD-independent NHEJ** (PMID: 42306942): *"RecA and Ku (but not LigD) are conditionally essential, which suggests that P. aeruginosa can conduct LigD-independent NHEJ."* This positions Ku as the central, most essential NHEJ factor in *Pseudomonas* — it remains functionally important even under conditions where the canonical ligase is dispensable, underscoring that Ku's end-recognition/bridging role is the indispensable core of the pathway.

### Finding 4 — Ku + LigD constitute a complete, self-sufficient two-component NHEJ machine; Ku provides end recognition/bridging while LigD provides all processing/ligation chemistry

The foundational reconstitution of bacterial NHEJ established that just two proteins suffice. Della et al. (2004, *Science*) demonstrated that prokaryotic Ku and ligase *"form a bona fide NHEJ system that encodes all the recognition, processing, and ligation activities required for DSB repair"* (PMID: 15499016). Within this division of labor, **Ku contributes end recognition and bridging**, whereas **LigD supplies all catalytic chemistry**. LigD is a modular enzyme comprising an N-terminal polymerase (POL) domain, a central 3′-phosphoesterase (PE) domain, and a C-terminal ATP-dependent ligase (LIG) domain (PMID: 23198659): *"LigD, a modular enzyme composed of a C-terminal ATP-dependent ligase domain (LIG), a central 3'-phosphoesterase domain (PE), and an N-terminal polymerase domain (POL)."*

The mutagenicity of NHEJ is mechanistically traceable to the LigD polymerase acting on Ku-held ends. LigD POL *"is proficient at adding templated and nontemplated deoxynucleotides and ribonucleotides to DNA ends in vitro and is the catalyst in vivo of unfaithful NHEJ events involving nontemplated single-nucleotide additions to blunt DSB ends"* (PMID: 23198659). Because Ku holds and synapses the ends while LigD performs these often-inaccurate additions and the final seal, the pathway is intrinsically error-prone — yet capable of joining even incompatible ends that other pathways cannot. This clarifies precisely which activities belong to Ku (non-catalytic end recognition, protection, bridging, LigD recruitment) versus its partner (all nucleotidyl-transfer and ligation chemistry).

### Finding 5 — Ku is the rate-limiting DNA-end sensor that protects broken ends from nucleolytic degradation and oligomerizes on DNA to drive synapsis

Detailed in vitro/in silico characterization of mycobacterial Ku demonstrates the physical properties underlying its end-protection and synapsis functions. Ku binds linear dsDNA with **positive cooperativity for substrates ≥40 bp**, can **slide along DNA**, and forms **DNA-dependent higher-order oligomers** implicated in synapsis; importantly, Ku binding both stabilizes the nucleoprotein complex and **shields DNA ends from exonuclease degradation** (PMID: 39122073): *"mKu's DNA binding stabilizes both the protein and DNA, while also shielding DNA ends from exonuclease degradation,"* and *"showing positive cooperativity for DNA* *substrates of 40 base pairs or longer, and its ability to slide along DNA strands."* The cooperative loading and sliding behavior provides the physical basis for the inward-threading and oligomerization steps described above.

Cryo-EM has now defined the Ku–Ku dimerization, DNA-binding, and synapsis interfaces and confirmed that **Ku is the rate-limiting factor** of two-component NHEJ, with NHEJ being the sole DSB-repair pathway available during mycobacterial dormancy (PMID: 41521670): *"relying on mycobacterial Ku (mKu) and ligase D, with mKu as the rate-limiting factor."* The broader pathway context is that NHEJ is one of several DSB-repair options — operating alongside homologous recombination (HR) and single-strand annealing (SSA) — and becomes especially important when a sister template is absent (PMID: 21219454). By analogy and by shared machinery, the *P. putida* Ku occupies the same rate-limiting, end-protecting position in its NHEJ system.

## Finding 6 — Ku's defining role is the "end-sensing / end-protection / end-tethering" anchor of NHEJ; in *P. putida* this underpins its use as a genome-engineering chassis

Contemporary mechanistic reviews frame NHEJ as five sequential phases — **end sensing, end protection, end tethering, end processing, and end ligation** — and emphasize that the pathway relies on protein assemblies rather than sequence complementarity (PMID: 41871908): *"relying instead on protein assemblies to bridge and stabilize the two DNA ends."* Critically, the end-tethering machineries are *"all anchored on Ku"* — placing Ku, the earliest-acting non-catalytic factor, at the organizing center of the first three phases. In the streamlined prokaryotic Ku+LigD system, Ku alone therefore carries out end sensing, end protection, and end tethering, before LigD performs processing and ligation.

In *P. putida* KT2440 specifically, the (atypical) presence of Ku+LigD makes NHEJ a practical lever in genome engineering. NHEJ is exploited alongside CRISPR-Cas9 and I-SceI double-strand-break tools; NHEJ status modulates the mutational outcomes of programmable DSBs and is an active consideration in the strain's editing toolbox (PMID: 36475478; PMID: 39031514). The I-SceI-based editing system for *P. putida* KT2440 uses a *"double-strand break introducing gene I-sceI and sacB counterselection marker"* (PMID: 39031514) — the applied context in which Ku-dependent end joining competes with (or is suppressed in favor of) homology-directed repair to achieve precise edits.

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## Mechanistic Model / Interpretation

The findings assemble into a coherent, well-supported model of Ku as the **non-catalytic scaffold** that initiates and organizes bacterial NHEJ in *P. putida*. The two-component machine can be summarized as a sequential hand-off:

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  Double-strand break in chromosomal DNA (cytoplasm / nucleoid)
                        |
                        v
 [1] END SENSING    Ku homodimer recognizes blunt/near-blunt dsDNA
                    ends (sequence-independent), loads as ring/clamp
                        |
                        v
 [2] END PROTECTION Ku shields ends from exonuclease degradation;
                    stabilizes the nucleoprotein complex
                        |
                        v
 [3] END TETHERING  Ku threads inward (basic C-terminal extension),
                    binds cooperatively (>=40 bp), slides, and
                    oligomerizes into arrays that BRIDGE / SYNAPSE
                    the two broken ends
                        |
                        v
 [4] LigD RECRUIT   Minimal C-terminal subregion of Ku recruits
                    and stimulates LigD
                        |
                        v
 [5] PROCESS+LIGATE LigD POL (fill-in, +/- nontemplated nt ->
                    mutagenic), LigD PE (3'-P processing),
                    LigD LIG (ATP-dependent seal)
                        |
                        v
       Repaired DNA (often with small insertions/deletions)
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**Division of labor.** The table below summarizes which activities are supplied by Ku versus LigD, clarifying that Ku is entirely non-catalytic and LigD supplies all chemistry.

| NHEJ phase | Protein responsible | Activity | Key evidence |
|---|---|---|---|
| End sensing | **Ku** (homodimer) | Sequence-independent dsDNA end binding | PMID: 26961308 |
| End protection | **Ku** | Shields ends from exonucleases; stabilizes complex | PMID: 39122073 |
| End tethering / synapsis | **Ku** | Threading, cooperative binding, oligomer arrays bridge two ends | PMID: 41118517; PMID: 41298423 |
| Ligase recruitment | **Ku** (minimal C-terminus) | Recruits/stimulates LigD | PMID: 26961308; PMID: 41298423 |
| End processing | **LigD** (POL, PE) | Fill-in synthesis (± nontemplated nt), 3′-phosphoesterase | PMID: 23198659 |
| End ligation | **LigD** (LIG) | ATP-dependent nick sealing | PMID: 15499016 |

**Localization.** Ku carries out its function **inside the cell, in the cytoplasm/nucleoid, directly at DSB sites on chromosomal DNA**. There is no signal peptide or transmembrane character implied by the family; the protein's substrate is the physical DNA end, so its site of action is defined by wherever a break occurs on the genome.

**Pathway logic.** Ku-dependent NHEJ is a **template-independent** and (in *Pseudomonas*) at least partly **RecA-independent** DSB-repair route. Because it does not require a homologous donor, it is the pathway of choice when a sister chromosome is unavailable — notably in **stationary phase / dormancy** — but this convenience comes at the cost of fidelity: LigD's ability to add nontemplated nucleotides to Ku-held ends makes the outcome intrinsically **mutagenic**. Ku is the **rate-limiting** component and the physical **anchor** for the whole assembly, which is why its loss cripples NHEJ even where LigD can be bypassed.

**Applied significance in *P. putida*.** Because KT2440 unusually possesses this machine, Ku-dependent NHEJ is a real force in the strain's genome-editing behavior: it competes with homology-directed repair at CRISPR-Cas9 and I-SceI breaks and biases outcomes toward small indels. Editing strategies for *P. putida* therefore either exploit NHEJ (for random small mutations) or suppress it (to favor precise, homology-directed edits).

## Evidence Base

| PMID | Organism / system | Contribution to this report | Type |
|---|---|---|---|
| 36475478 | *P. putida* KT2440 | **Direct**: KT2440 encodes Ku+LigD; NHEJ repairs Cas9 DSBs; intrinsically mutagenic | Primary (in vivo, target organism) |
| 42306942 | *P. aeruginosa* | Ku (not LigD) conditionally essential; LigD-independent NHEJ; Ku is central | Primary (genetics, close relative) |
| 26961308 | Bacterial Ku | C-terminus dissection: LigD interaction vs. threading/bridging roles | Primary (biochemistry) |
| 41118517 | *B. subtilis* | Cryo-EM (2.74 Å): Ku homodimer arrays bridge two blunt DNA ends | Primary (structure) |
| 41298423 | *M. tuberculosis* | Cryo-EM: Ku proteo-filament; C-terminus controls loading + LigD recruitment | Primary (structure) |
| 39122073 | Mycobacterial Ku | Cooperative binding (≥40 bp), sliding, oligomerization, exonuclease shielding | Primary (in vitro/in silico) |
| 41521670 | Mycobacterial Ku | Cryo-EM of Ku–DNA synaptic complex; Ku is rate-limiting; NHEJ in dormancy | Primary (structure) |
| 15499016 | Prokaryotic Ku+LigD | Foundational: Ku+LigD = complete two-component NHEJ machine | Primary (reconstitution) |
| 23198659 | *M. smegmatis* | LigD domain architecture (POL/PE/LIG); POL is catalyst of mutagenic additions | Primary (biochemistry) |
| 21219454 | *M. smegmatis* | NHEJ is one of three DSB pathways (HR, NHEJ, SSA); error-prone | Primary (genetics) |
| 41871908 | Review | Five-phase NHEJ framework; end-tethering "all anchored on Ku" | Review |
| 39031514 | *P. putida* KT2440 | I-SceI DSB-based genome editing — applied context for NHEJ | Primary (methods) |

**How the evidence supports the annotation.** The target-organism paper (PMID: 36475478) directly establishes that *P. putida* KT2440 possesses and uses Ku for NHEJ, and that the pathway is mutagenic in vivo. The reconstitution paper (PMID: 15499016) demonstrates that Ku+LigD alone suffice for the entire pathway, defining the minimal machine. Structural studies in *B. subtilis*, *M. tuberculosis*, and *Mycobacterium* (PMID: 41118517; PMID: 41298423; PMID: 41521670) provide the physical basis for end bridging, synapsis, and rate-limiting behavior. Biochemical dissection (PMID: 26961308; PMID: 39122073) assigns specific functions to Ku subregions and confirms end protection. The LigD papers (PMID: 23198659; PMID: 21219454) delineate the boundary between Ku's non-catalytic role and LigD's chemistry.

**Note on cross-organism inference.** Because there is currently no published high-resolution structure or detailed biochemistry of the *P. putida* KT2440 Ku protein itself, several mechanistic details (threading, oligomerization, synapsis interfaces, exonuclease protection) are inferred from well-characterized bacterial orthologs (*B. subtilis*, *M. tuberculosis/smegmatis*). This inference is strongly justified by the high conservation of the prokaryotic Ku family (shared Pfam/InterPro domains) and by the direct in vivo demonstration that KT2440 Ku participates in NHEJ. The applied *P. putida* observations (PMID: 36475478; PMID: 39031514) anchor the general model to the target organism.

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## Limitations and Knowledge Gaps

1. **No target-specific structure or biochemistry.** The detailed mechanistic attributes (DNA sliding, cooperative loading ≥40 bp, oligomer-array synapsis, exonuclease shielding, precise C-terminal subregion functions) are established for mycobacterial and *B. subtilis* Ku, not for *P. putida* PP_3255. While family conservation makes transfer of these properties well-founded, the KT2440 protein has not been crystallized, cryo-EM-solved, or biochemically reconstituted in isolation.

2. **LigD-independent NHEJ is incompletely defined.** The *P. aeruginosa* finding that Ku (but not LigD) is conditionally essential implies an alternative ligase or LigD-independent route (PMID: 42306942), but the identity of the substitute ligase and whether the same holds in *P. putida* remains unresolved.

3. **Physiological trigger conditions in *P. putida* are inferred, not measured.** The claim that NHEJ dominates in stationary phase/dormancy derives largely from mycobacterial work (PMID: 41521670; PMID: 21219454). The specific growth-phase and stress dependence of *ku* expression and NHEJ activity in KT2440 has not been directly quantified here.

4. **Quantitative parameters absent for the target.** DNA-binding affinities (Kd), stoichiometry, and end-joining efficiencies are available for orthologs but not for Q88HU8 specifically. Effect sizes for how *ku* deletion/overexpression shifts edit outcomes in KT2440 are described qualitatively in the literature rather than tabulated here.
5. **Regulation and interactome unexplored.** Whether *P. putida* Ku is regulated (transcriptionally, post-translationally) or interacts with additional accessory factors beyond LigD (as eukaryotic Ku does with XLF, XRCC4, etc.) is not established for this organism.

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## Proposed Follow-up Experiments / Actions

1. **Structural determination of *P. putida* Ku.** Solve the cryo-EM/crystal structure of PP_3255 alone and in complex with DNA (and with LigD), to confirm the homodimer clamp, threading, and synapsis interfaces directly in the target protein rather than by homology.
2. **In vitro reconstitution with KT2440 proteins.** Purify PP_3255 Ku and *P. putida* LigD and reconstitute NHEJ on defined blunt and non-complementary end substrates; measure DNA-binding affinity, cooperativity (test the ≥40 bp threshold), exonuclease-protection, and end-joining fidelity/efficiency. Compare to mycobacterial benchmarks.
3. **Targeted C-terminal mutagenesis.** Recreate the minimal-C-terminus vs. basic-extension dissection (PMID: 26961308) in the KT2440 protein to test whether LigD-recruitment and threading/bridging map to the same subregions.
4. **Growth-phase and stress profiling.** Quantify *ku* (PP_3255) and *ligD* expression across exponential, stationary, and stress (ionizing radiation, desiccation, oxidative, antibiotic) conditions in KT2440, and correlate with NHEJ-mediated repair efficiency using an inducible DSB reporter.
5. **Editing-outcome quantification.** Systematically measure how Δ*ku*, WT, and *ku*-overexpression backgrounds change the indel spectrum and precise-edit efficiency at CRISPR-Cas9 and I-SceI breaks in KT2440, to establish quantitative rules for exploiting or suppressing NHEJ during genome engineering (PMID: 36475478; PMID: 39031514).
6. **Test for LigD-independent NHEJ in *P. putida*.** Replicate the *P. aeruginosa* conditional-essentiality experiment (PMID: 42306942) in KT2440 and identify any alternative ligase that can act with Ku.

## Conclusion

The gene *ku* (PP_3255, UniProt Q88HU8) of *Pseudomonas putida* KT2440 encodes the **prokaryotic Non-Homologous End Joining protein Ku**, a non-catalytic, homodimeric DNA-double-strand-break end-binding scaffold. Its primary function is to **sense, protect, and tether** broken chromosomal DNA ends in a sequence-independent manner, then recruit the catalytic ligase LigD to complete repair. Ku is the rate-limiting anchor of the minimal two-component (Ku+LigD) bacterial NHEJ machine, acts in the cytoplasm/nucleoid at DSB sites, and provides a template-independent, intrinsically mutagenic repair route that is biologically most relevant when no homologous template is available — and that is a practical determinant of genome-editing outcomes in this biotechnologically important strain. The gene symbol and family assignment are unambiguous and fully consistent with the UniProt annotation.

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